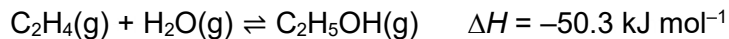


12 Ethanol is produced industrially by reacting ethene and steam.



K_p has a value of 1.8×10^{-5} and the partial pressures of the reactants at equilibrium are shown.

reactant	partial pressure / $\times 10^6 \text{ Pa}$
ethene	4.8
steam	2.8

Which statement is correct?

- A** Adding a catalyst increases the value of K_p for the reaction at equilibrium.
- B** The overall process is a nucleophilic addition reaction.
- C** Increasing the temperature will increase the K_p for the reaction.
- D** Partial pressure of ethanol at equilibrium is $242 \times 10^6 \text{ Pa}$.